

TREMORWATCH

AI-Powered Seismic Risk Intelligence for Disaster Management & Emergency Response

POSTER BATTLE • 15 SEPT 2026 • ACM STUDENT CHAPTER, SVKM IOT DHULE

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ABSTRACT

TREMORWATCH turns the live USGS global feed into emergency-response intelligence. From **11,143 events in 30 days** it validates the catalog against Gutenberg-Richter ($b = 1.15$), maps 5°-grid risk zones where significant ($M \geq 5.0$) events concentrate, and triages every newly detected event with a Random Forest model. Under strict walk-forward cross-validation (**8,912 unseen future events, zero leakage**), flagged events are **13.3× more likely** to be significant than a random alert, and the system catches **79%** of significant events — faster, better-informed response.

PROCESS DIAGRAM



OBJECTIVES

- **Fast triage** — rank detected events by response priority in under one second
- **Risk zoning** — decide where to pre-position teams, caches and drills
- **Situational awareness** — one live picture of global seismic activity and severity
- **Trustworthy evaluation** — leak-free validation responders can rely on

PROBLEM STATEMENT

Earthquakes cannot be predicted with certainty — but their **where** and **likelihood** are strongly patterned, and response is won or lost in the first minutes. When a significant event is detected, responders must answer immediately: **how severe is it, which events deserve action, and where are we already prepared?**

Without structured data intelligence, alert overload and guesswork delay those answers. TREMORWATCH converts a public seismic stream into them — continuously, at machine speed.

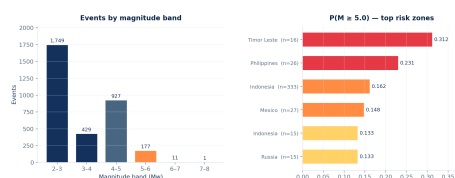
INNOVATION

- An **honest risk layer**, not a false "prediction" — posterior risk zones plus per-event ML triage
- **Walk-forward evaluation** on 8,912 unseen future events — zero leakage, the standard most posters skip
- The model independently learned the **Ring of Fire** — location + depth carry ~90% of the signal
- **13.3× lift** over the 1.5% base rate, **79% catch** of significant events — misses shown on purpose

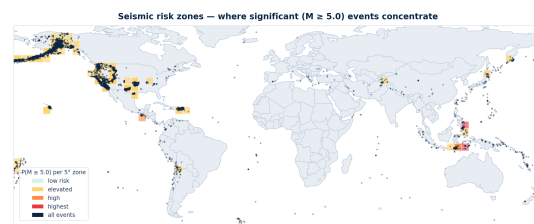
METHODOLOGY

- **Data** — USGS global feed (public GeoJSON, no API key); 13 Aug – 12 Sept 2026; 11,143 events
- **Validation** — Gutenberg-Richter fit $b = 1.15$ (expected ≈ 1.0); catalog health confirmed
- **Risk zones** — 5° grid; $P(M \geq 5.0)$; minimum 15 events per zone for stability
- **Model** — Random Forest, 300 trees, balanced classes; features known at detection: grid location, depth, hour, day
- **Evaluation** — 4 sequential hold-out windows; threshold 0.30; precision / recall / lift

SURVEY



PROPOSED MODEL



Every detected event scored in < 1 s; only top-confidence, in-risk-zone events page on-call responders — cutting alert noise while catching **79%** of significant events.

CONCLUSION

Seismic response can be faster and smarter. Combining observed risk zones with leak-free ML triage, TREMORWATCH turns a public data stream into actionable priorities — resources pre-positioned where risk concentrates, responders paged only for events that matter. The same **zone → triage → alert** architecture generalises to cyclones, floods and any hazard with a live data feed.

REFERENCES

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- 3 Pedregosa, F. et al. (2011). Scikit-learn: Machine learning in Python. JMLR 12: 2825–2830
- 4 Aki, K. (1965). ML estimate of b in $\log N = a - bM$. Bull. Seismol. Soc. Am. 55(2): 73–87